

# Munkres 6

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## 1 Nagata-Smirnov Metrization Theorem

Let  $\mathcal{A}$  be an open covering of the metric space  $X$ . There exists a countably locally finite open covering  $\mathcal{B}$  refining  $\mathcal{A}$ .

Give  $\mathcal{A}$  a well-order. Given open  $U \in \mathcal{A}$ , define the “contraction”  $S_n(U) = \{x \in U : B(x, 1/n) \subset U\}$ . Let  $T_n(U_\alpha) = S_n(U_\alpha) - \bigcup_{\beta < \alpha} U_\beta$ . Finally, let  $W_n(U) = B(T_n, 1/(3n))$ . Let  $\mathcal{B}_n = \{W_n(U) : U \in \mathcal{A}\}$ . I claim  $\mathcal{B} = \bigcup_{n < \omega} \mathcal{B}_n$  is our desired covering.

First, if  $x \in X$ , there exists  $\alpha \in \text{Ord}$  for which  $x \in A_\alpha \in \mathcal{A}$ . Take  $\alpha$  to be minimal. Since  $A_\alpha$  is open, pick  $n \in \mathbb{N}$  for which  $B(x, 1/n) \subset A_\alpha$ , so  $x \in S_n(U_\alpha)$ , so  $x \in T_n(U_\alpha) \subset W_n(U_\alpha)$ , so  $\mathcal{B}$  covers  $X$ .

I claim  $\mathcal{B}_n$  is locally finite. I claim first for  $\alpha < \beta$  that  $d(T_n(U_\alpha), T_n(U_\beta)) \geq 1/n$ . Note  $T_n(U_\beta) \subset U_\beta - U_\alpha$  and  $T_n(U_\alpha) \subset S_n(U_\alpha)$ . Sps FTSOC  $d(T_n(U_\alpha), T_n(U_\beta)) < 1/n$ . Then there exist  $x \in T_n(U_\alpha)$  and  $y \in T_n(U_\beta)$  such that  $d(x, y) < 1/n$ , so  $B(x, 1/n) \not\subset U_\alpha \rightarrow x \notin S_n(U_\alpha)$ , cont.

Sps FTSOC  $B(x, 1/(6n))$  intersects  $W_n(U), W_n(V)$  for  $U \neq V$ . Then there exist  $y_1 \in U, y_2 \in V$  such that  $d(x, y_1) < 1/(6n)$  and  $d(x, y_2) < 1/(6n)$ , so  $d(y_1, y_2) \leq d(y_1, x) + d(x, y_2) < 1/(3n)$ , cont. as  $d(W_n(U), W_n(V)) \geq 1/(3n)$ .

Let  $X$  be second-countable. Show that a collection of subsets  $\mathcal{A}$  is countably locally finite iff  $\mathcal{A}$  is countable.

If  $\mathcal{A}$  is countable, then  $\mathcal{A}$  is countably locally finite (one set per  $\mathcal{A}_n$ ). Suppose  $\mathcal{A} = \bigcup_{n < \omega} \mathcal{A}_n$  is countably locally finite. Consider a (locally finite) element  $\mathcal{A}_n$  and take  $\emptyset \notin \mathcal{A}_n$ . For all  $x \in X$ , pick  $B(x)$  intersecting finitely many  $A \in \mathcal{A}_n$ . Denote the set of these  $B(x)$  by  $\mathcal{B}' \subset \mathcal{B}$ . Denote by  $I = \{(A, B) \in \mathcal{A}_n \times \mathcal{B}' : A \cap B \neq \emptyset\}$ .

If  $A \in \mathcal{A}_n$  is nonempty, let  $x \in A$  and  $x \in B$  for some  $B \in \mathcal{B}'$ . Then  $(A, B) \in I$ , so  $|\mathcal{A}_n| \leq |I|$ . Further,  $I = \bigcup_{B \in \mathcal{B}'} \{(A, B) \in I\}$ , which is a countable union of finite sets. Hence,  $\mathcal{A}_n$  is countable, and so is  $\mathcal{A}$ .

If  $X$  is metrizable, then  $X$  has a countably locally finite basis. Let  $\mathcal{A}_n$  be a covering of  $X$  with  $B(x, 1/n)$ . Pick  $\mathcal{B}_n$  to be a countably locally finite open covering of  $X$  refining  $\mathcal{A}_n$ . Further, (by before) take the elements of  $\mathcal{B}_n$  to have diameter at most  $2/n$ . Let  $\mathcal{B} = \bigcup_{n < \omega} \mathcal{B}_n$ , a countably locally finite open covering of  $X$ .

If  $x \in U \subset X$ , pick  $B(x, 1/n) \subset U$ . Since  $\mathcal{B}_{2n}$  covers  $X$ , pick  $B(x) \in \mathcal{B}_{2n}$ . Since  $B$  has a diameter of at most  $1/n$ ,  $B \subset B(x, 1/n) \subset U$ , so  $\mathcal{B}$  is indeed a basis.

If  $X$  is regular and has a countably locally finite basis, then  $X$  is normal. Let  $\mathcal{B} = \bigcup_{n < \omega} \mathcal{B}_n$  be such a basis,  $A, B \subset X$  be disjoint closed. Let  $a \in A$  and pick  $N(a) \cap B = \emptyset$  and by regularity pick  $\overline{U(x)} \subset N(a)$  for  $U \in \mathcal{B}$ .  $U \in \mathcal{B}_n$  for some  $n \in N$ . Put  $x \in A_n$ .

Let  $n \in N$ . Consider the open coverings  $\{U_\alpha\}_{\alpha \in J}$  and  $\{V_\alpha\}_{\alpha \in J}$  (if indexed by different sets, allow duplicates) of  $A_n, B_n$ , resp st  $\overline{U} \cap B = \overline{V} \cap A = \emptyset$  for all  $U, V$ . Let  $U'_\alpha = U_\alpha - \bigcup_{\beta \leq \alpha} \overline{V_\beta}$ . Since  $\{V_\alpha\} \subset \mathcal{B}_n$ ,  $\{V_\alpha\}$  is locally finite. Then  $\bigcup_{\beta \leq \alpha} \overline{V_\beta} = \overline{\bigcup_{\beta \leq \alpha} V_\beta}$  is closed and  $U'_\alpha$  is open.  $U'_\alpha \cap A = U_\alpha \cap A$ , so  $\{U'_\alpha\}$  covers  $A_n$ . Construct  $\{V'_\alpha\}$  similarly.

Let  $U_n = \bigcup_{\alpha \in J} U'_\alpha$  and define  $V_n$  similarly. Sps FTSOC  $U_n \cap V_n \neq \emptyset$ . Then  $U'_\alpha \cap V'_\beta \neq \emptyset$  for some  $\alpha, \beta$  and WLOG  $\alpha \leq \beta$ . However,  $V'_\beta \cap U_\alpha = \emptyset$  and  $U'_\alpha \subset U_\alpha$ , so  $U'_\alpha \cap V'_\beta = \emptyset$ , cont.

Now  $\{U_n\}_{n \in N}, \{V_n\}_{n \in N}$  cover  $A, B$ , resp. since if  $x \in A$ , then  $x \in A_n$  for some  $n$ , so  $x \in U_n$ . We also have  $\overline{U_n} \cap B = \overline{V_n} \cap A = \emptyset$  for all  $n \in N$ : since  $\{U_\alpha\}$  is locally finite,  $\overline{U_n} = \overline{\bigcup_{\alpha \in J} U_\alpha} = \bigcup_{\alpha \in J} \overline{U_\alpha}$ , which is disjoint to  $B$ . Then define similarly,  $U'_n = U_n - \bigcup_{k \leq n} \overline{V'_k}$  and  $V'_n$ . By similar reasoning,  $U = \bigcup_{n < \omega} U'_n$  and  $V = \bigcup_{n < \omega} V'_n$  are disjoint open covering  $A, B$ .

Munkres Lemma: Let  $X$  be regular with a countably locally finite basis  $\mathcal{B}$ . Then for all open  $U \subset X$ , there exists a countable collection  $\{U_n\}$  such that  $U = \bigcup_{n < \omega} U_n = \bigcup_{n < \omega} \overline{U_n}$ .

Let  $\mathcal{B} = \bigcup_{n < \omega} \mathcal{B}_n$ , where  $\mathcal{B}_n$  is locally finite. Denote by  $\mathcal{B}'$  the set of basis elements  $B \in \mathcal{B}$  such that  $\overline{B} \subset U$ . Sort  $\mathcal{B}'$  into  $\{\mathcal{B}'_n\}$ , i.e.  $\mathcal{B}'_n = \mathcal{B}' \cap \mathcal{B}_n$ . Let  $U_n = \bigcup \mathcal{B}'_n$ . I claim that these  $\{U_n\}$  work.

First, since  $\mathcal{B}'_n$  is locally finite,  $\overline{U_n} = \bigcup_{B \in \mathcal{B}'_n} \overline{B} \subset U$ , so  $\bigcup_{n < \omega} \overline{U_n} \subset U$ . Further, if  $x \in U$ , by regularity, pick  $\overline{V(x)} \subset U$ . Pick basis  $B(x) \subset V$ , so  $\overline{B} \subset U$  and  $B \in \mathcal{B}'$ . It follows for some  $n$ ,  $x \in U_n$ , so  $x \in \bigcup_{n < \omega} U_n$ . Clearly,  $\bigcup_{n < \omega} U_n \subset \bigcup_{n < \omega} \overline{U_n}$ , so we have for some sets  $A, B, C$  that  $A \subset B \subset C \subset A$ ,

so  $A = B = C$ .

Second Munkres hint: any closed  $A \subset X$  is a  $G_\delta$  set. Proof:  $X - A = \bigcup_{n < \omega} \overline{U_n}$  is open, so  $A = X - \bigcup_{n < \omega} \overline{U_n} = \bigcap_{n < \omega} (X - \overline{U_n})$ .

If  $X$  is perfectly normal and countably locally finite, then  $X$  is metrizable.

Suppose  $\mathcal{B} = \bigcup_{n < \omega} \mathcal{B}_n$ . Let  $B \in \mathcal{B}$ . For all  $n \in \mathbb{N}$  such that  $B \in \mathcal{B}_n$ , construct  $f_\alpha : X \rightarrow [0, 1/n]$  such that  $f_\alpha$  vanishes exactly on  $X - B$  (since  $X$  is perfectly normal). Denote by  $J$  the set of such  $\alpha$  and let  $\mathbb{R}^J$  have the uniform metric topology. Define  $F : X \rightarrow \mathbb{R}^J$  by  $F(x) = (f_\alpha(x))_{\alpha \in J}$ . I claim that  $F$  is an embedding.

I claim  $F$  is continuous. Let  $x \in X$ ,  $y = F(x)$ , and  $T = \prod_{\alpha \in J} (y_\alpha - \epsilon, y_\alpha + \epsilon)$  be a tube. Pick  $n_0$  such that  $1/n_0 < \epsilon$ . Let  $n < n_0$ . Pick  $U_n(x)$  intersecting finitely many elements of  $\mathcal{B}_n$  and for each such element let  $f_\alpha$  be associated. Define  $V_n = \bigcap f_\alpha^{-1}((y_\alpha - \epsilon, y_\alpha + \epsilon))$ , which is open. Now let  $V = \bigcap_{n < n_0} V_n$  and  $U = \bigcap_{n < n_0} U_n$ . I claim  $F(U \cap V) \subset T$ .

If  $f_\alpha$  is associated with  $n \geq n_0$ , then two values in the range of  $f_\alpha$  differ by at most  $1/n$ , so  $f_\alpha(X) \subset [y_\alpha - 1/n, y_\alpha + 1/n] \subset (y_\alpha - \epsilon, y_\alpha + \epsilon)$ . Else,  $f_\alpha$  is associated with  $n < n_0$  and  $B \in \mathcal{B}_n$ . If  $B$  is disjoint to  $U_n$ , then  $f_\alpha(U \cap V) \subset f_\alpha(U_n) = \{0\}$ . Else,  $B$  intersects  $U_n$  and  $f_\alpha(U \cap V) \subset f_\alpha(V_n) \subset (y_\alpha - \epsilon, y_\alpha + \epsilon)$ .

$F$  is injective. If  $x \neq y$ , pick  $B(x)$  disjoint to  $y$ . Then for some  $\alpha$ ,  $f_\alpha(x) > 0$  and  $f_\alpha(y) = 0$ .

$F$  restricts to a homeomorphism. Let  $U(x) \subset X$  be open. Pick  $B(x) \subset U$ , so for some  $\alpha$ ,  $f_\alpha(X - B) = \{0\}$ . Then  $F^{-1}(\pi_\alpha^{-1}(\mathbb{R}^+)) = (\pi_\alpha \circ F)^{-1}(\mathbb{R}^+) = f_\alpha^{-1}(\mathbb{R}^+) \subset B \subset U$ , i.e., for all  $U(x) \subset X$  open, there exists  $W(y) \subset F(X)$  open for which  $F^{-1}(W) \subset U$ .

$A \subset X$  is a  $F_\sigma$  set iff  $B = X - A$  is a  $G_\delta$  set.

Let  $A = \bigcup_{n < \omega} C_n$  be  $F_\sigma$ , so  $X - A = X - \bigcup_{n < \omega} C_n = \bigcap_{n < \omega} (X - C_n)$ , so  $X - A$  is  $G_\delta$ . Conversely, if  $B = \bigcap_{n < \omega} U_n$  is  $G_\delta$ ,  $X - B = X - \bigcap_{n < \omega} U_n = \bigcup_{n < \omega} (X - U_n)$  is  $F_\sigma$ .

Let  $X$  be  $T_1$  with a locally finite basis. Let  $x \in X$ , so the neighborhood basis  $\mathcal{N}_x$  is also locally finite. Pick  $U(x)$  intersecting finitely many elements of  $\mathcal{N}_x$ . Since  $x \in U(x)$ ,  $U(x)$  intersects every element of  $\mathcal{N}_x$ , so  $\mathcal{N}_x$  is finite. Since  $\{x\} = \bigcap \mathcal{N}_x$  is a finite intersection,  $\{x\}$  is open, so  $X$  is discrete.

Consider  $\mathbb{R}^J$  with the uniform topology and  $J$  being uncountable.  $\mathbb{R}^J$  has an uncountable basis since  $\mathbb{Z}^J \subset \mathbb{R}^J$  is uncountable discrete (as a subspace).  $\mathbb{R}^J$  is

metrizable and therefore has a countable locally finite basis.

Every open cover  $\mathcal{A}$  of  $X$  has a countably locally discrete cover refining  $\mathcal{A}$  covering  $X$ . A countable union of countably locally discrete covers is countably locally discrete. If  $X$  is metrizable,  $X$  has a countably locally discrete basis. If  $X$  has such a basis, that basis is countably locally finite.

## 2 Paracompactness

Attempt on the paracompactness (or not) of  $\omega_1$ . For all ordinals  $\alpha > 0$ , there exist unique  $\beta, \gamma$  such that  $\alpha = \omega^\beta \gamma$ , where  $\gamma \notin \text{lOrd}$ .

(1) Existence. Write  $\alpha = \omega^{\beta_1} k_1 + \omega^{\beta_2} k_2 + \dots + \omega^{\beta_n} k_n$  in normal form. For each  $i \in [n-1]$ , since  $\beta_i \geq \beta_n$ , there exists  $\delta_i$  such that  $\beta_i = \beta_n + \delta_i$ , so

$$\begin{aligned} \alpha &= \omega^{\beta_n + \delta_1} k_1 + \omega^{\beta_n + \delta_2} k_2 + \dots + \omega^{\beta_n + \delta_{n-1}} k_{n-1} + \omega^{\beta_n} k_n \\ &= \omega^{\beta_n} (\omega^{\delta_1} k_1 + \omega^{\delta_2} k_2 + \dots + \omega^{\delta_{n-1}} k_{n-1} + k_n) \\ &= \omega^{\beta_n} \gamma. \end{aligned}$$

(2) Uniqueness. Suppose  $\alpha = \omega^\beta \gamma = \omega^{\beta'} \gamma'$ . WLOG  $\beta < \beta'$  and let  $\beta' = \beta + \delta$  for  $\delta > 0$ . Then  $\omega^\beta \gamma = \omega^{\beta + \delta} \gamma' = \omega^\beta (\omega^\delta \gamma')$ . Since multiplication is strictly increasing in the second operand with nonzero first operand,  $\gamma = \omega^\delta \gamma'$ , cont, as  $\gamma \notin \text{lOrd}$  (if  $\gamma' \in \text{lOrd}$  or  $\gamma' = \gamma'' + 1$ ,  $\gamma \in \text{lOrd}$ ).

For  $\beta < \omega_1$ , define  $X_\beta = \{\omega^\beta \gamma : \gamma < \omega_1\} \subset \omega_1$ . Note that the  $\{X_\beta\}$ 's form a decreasing chain: if  $\beta' < \beta$ , let  $\alpha \in X_\beta$ , so  $\alpha = \omega^\beta \gamma$ . Let  $\beta = \beta' + \delta$ , so  $\alpha = \omega^{\beta' + \delta} \gamma = \omega^{\beta'} (\omega^\delta \gamma) \in X_{\beta'}$ .

Let  $0 < \alpha < \omega_1$ . Pick  $\beta$  to be maximal such that  $\alpha \in X_\beta$ . Then there exists  $\alpha' < \alpha$  for which  $(\alpha', \alpha] \cap X_\beta = \{\alpha\}$ . Sps FTSOC  $(\alpha', \alpha) \cap X_\beta \neq \emptyset$  for all  $\alpha' < \alpha$ . Then  $\alpha$  is a limit point of  $X_\beta$ , or being in  $X_{\beta+1}^*$ , cont max of  $\beta$ .

\*Let  $A = X_\beta(\alpha)$ . Sps FTSOC  $A$  has a maximum, say  $\delta$ . Then  $(\delta, \alpha) \cap X_\beta = \emptyset$ , cont. Hence,  $A$  has no maximum, and  $\text{type}(A) \in \text{lOrd}$ . Then  $\alpha = \sup A = \lim_{\gamma \rightarrow \text{type}(A)} \omega^\beta \gamma = \omega^\beta \text{type}(A) \in X_{\beta+1}$ . \*logic seems suspicious, but result is fine.

If  $X$  is paracompact Hausdorff,  $X$  is normal. First, we show  $X$  is regular. Let  $A \subset X$  be closed and  $x \notin A$ . Since  $X$  is Hausdorff, for all  $a \in A$ , pick  $U(x), V(a)$  disjoint. Since  $X$  is paracompact and  $\{V(a)\}, X - A$  cover  $X$ , pick locally finite open refinement  $\{V'\}, \mathcal{E}$ , where the elements of  $\mathcal{E}$  refine  $X - A$  and  $\{V'\}$  covers  $A$ . Since  $\{V'\}$  is locally finite, pick  $W(x)$  intersecting finitely many  $V'$ 's. Let the finite corresponding set of  $U$ 's be  $\mathcal{U}$ , so the neighborhood  $W \cap \bigcap \mathcal{U}$  of  $x$  is disjoint to  $\bigcup V'$ .

$X$  is normal. Let  $A$  be closed and  $U$  be open such that  $A \subset U$ . By regularity, for all  $a \in A$ , pick  $\overline{V(a)} \subset U$ . Pick locally finite open refinement  $\{V'\}$  covering  $A$ . Then  $\bigcup V' = \bigcup \overline{V'} \subset U$ , so we have found  $\overline{V(A)} \subset U$ .

Book proof of part I: Let  $A$  be closed and  $x \notin A$ . For all  $a \in A$ , pick  $\overline{V(a)}$  disjoint to  $x$ . Pick locally finite open refinement  $\{V'\}$  covering  $A$ . Then  $\bigcup V' = \bigcup \overline{V'} \subset \bigcup \overline{V}$  is disjoint to  $x$ .

Let  $X$  be regular. Then the following conditions on  $X$  are equivalent: every open covering of  $X$  has a refinement that is:

- (1) an open covering of  $X$  and countably locally finite.
- (2) a covering of  $X$  and locally finite.
- (3) a closed covering of  $X$  and locally finite.
- (4) an open covering of  $X$  and locally finite.

(1)  $\longrightarrow$  (2). Define  $\mathcal{B}'_n = \{B' = B - \bigcup_{k < n} \bigcup \mathcal{B}_k : B \in \mathcal{B}_n\}$ , so  $\mathcal{B}' = \bigcup_{n < \omega} \mathcal{B}'_n$  is a covering of  $X$  since  $\bigcup_{k \leq n} \bigcup \mathcal{B}'_k = \bigcup_{k \leq n} \bigcup \mathcal{B}_k$ . If  $x \in X$ , pick  $n < \omega$  such that  $B(x) \in \mathcal{B}_n$ . For all  $k \leq n$ , since  $\mathcal{B}_k$  is locally finite, pick  $U_k(x)$  intersecting finitely many elements of  $\mathcal{B}_k$ . Then  $U(x) = B \cap \bigcap_{k \leq n} U_k(x)$  intersects finitely many elements of  $\bigcup_{k \leq n} \mathcal{B}_k$ . For  $k > n$ , any element of  $\mathcal{B}'_k$  is disjoint to  $B$ , so  $U$  intersects finitely many elements of  $\mathcal{B}'$ .

\* $\mathcal{B}'$  is a refinement of  $\mathcal{B}$  since  $B' \subset B$ .

(2)  $\longrightarrow$  (3). Suppose  $\mathcal{A}$  is a covering of  $X$ . For all  $U \in \mathcal{A}$  and  $x \in U$ , pick by regularity  $\overline{V(x)} \subset U$ , so  $\{V\}$  is an open covering of  $X$ . Pick a locally finite covering  $\{V'\}$  of  $\{V\}$ . Then  $\{\overline{V'}\}$  is a locally finite closed covering of  $X$  refining  $\mathcal{A}$ .

(3)  $\longrightarrow$  (4). Let  $\mathcal{B}$  be any locally finite covering refining  $\mathcal{A}$  covering  $X$ . Since  $\mathcal{B}$  is locally finite, for all  $x \in X$ , pick  $U(x)$  intersecting finitely many elements of  $\mathcal{B}$ , so  $\{U\}$  covers  $X$ . Pick a locally finite closed refinement  $\mathcal{C}$  of  $\{U\}$  covering  $X$ . Note that each element of  $\mathcal{C}$  intersects finitely many elements of  $\mathcal{B}$ .

Let  $B \in \mathcal{B}$ . Define  $V(B) = X - \bigcup \{C : C \in \mathcal{C}, C \cap B = \emptyset\}$  ( $V(B)$  is a neighborhood of  $B$ ). I claim  $\{V\}$  is a locally finite open covering of  $X$  (not necessarily refining  $\mathcal{A}$ ).

Let  $x \in X$ . Since  $\mathcal{C}$  is locally finite, pick  $W(x)$  intersecting finitely many elements of  $\mathcal{C}$  and denote this subset by  $\mathcal{D}$ . Suppose  $W \cap V(B) \neq \emptyset$  for some  $B \in \mathcal{B}$ . Let  $y \in W \cap V(B)$ , and since  $\mathcal{C}$  covers  $X$ , there exists  $D \in \mathcal{D}$  such that  $y \in D$ . Further,  $D \cap B \neq \emptyset$  as otherwise  $y \notin V(B)$ .

Sps FTSOC there exists an infinite set of  $\mathcal{B}' \subset \mathcal{B}$  such that for all  $B \in \mathcal{B}'$  we have a map  $f : \mathcal{B}' \rightarrow \mathcal{D}$  such that  $B \cap f(B) \neq \emptyset$ . By Pigeonhole principle, there exists  $D \in \mathcal{D}$  such that  $f^{-1}(D)$  is infinite. That is, there exist infinitely many elements of  $\mathcal{B}$  intersecting an element in  $\mathcal{C}$ , cont. Hence,  $W \cap V(B) \neq \emptyset$  for only finitely many elements of  $\mathcal{B}$ .

Now, since  $\mathcal{B}$  refines  $\mathcal{A}$ , for each  $V(B)$ , pick  $B \subset U$  and let  $\{V(B) \cap U\}$  be our locally finite open covering of  $X$  refining  $\mathcal{A}$ .

(4)  $\longrightarrow$  (1) is trivial.

Munkres proof for (3)  $\longrightarrow$  (4) (though my proof was hinted by his). Pick  $W(x)$  intersecting finitely many elements of  $\mathcal{C}$ .  $\mathcal{C}$  covers  $X$ , so let  $W(x)$  be covered only by  $C_1, C_2, \dots, C_n$ . It suffices to show  $C_i$  intersects finitely many  $V(B)$ . If  $C_i \cap V(B) \neq \emptyset$ , then  $C_i \cap B \neq \emptyset$ . If  $C_i$  intersects infinitely many  $V(B)$ , then  $C_i$  intersects infinitely many  $B$ , cont.

Corollary: every metrizable space is paracompact. If  $X$  is metrizable, every open cover has a countably locally finite refinement covering  $X$ . Since  $X$  is also regular, for every open covering  $X$  has a locally finite open refinement covering  $X$ .

Corollary: every regular Lindelof space is paracompact. Let  $\mathcal{A}$  be an open covering of  $X$ . Let  $\mathcal{A}' \subset \mathcal{A}$  be a countable covering of  $X$ , so  $\mathcal{A}'$  is countably locally finite. Then again, since  $X$  is regular, for every covering  $\mathcal{A}$  of  $X$ , there exists a locally finite open refinement covering  $X$ .

The product of two paracompact spaces need not be paracompact.  $\mathbb{R}_l$  is regular Lindelof and therefore paracompact.  $\mathbb{R}_l^2$  is Hausdorff and not normal.  $\mathbb{R}_K$  is not paracompact since it is Hausdorff but not normal (in fact not regular).  $\mathbb{R}^J$  is not paracompact for  $J$  uncountable since  $\mathbb{R}^J$  is regular but not normal.

Let  $\mathcal{A}$  be an open covering of  $X$ . Let  $\{\phi_\alpha\}_{\alpha \in J}$  be a collection of functions  $\phi : X \rightarrow I$  such that  $\mathcal{B} = \{\text{support } \phi_\alpha : \alpha \in J\}$  is a locally finite closed refinement of  $\mathcal{A}$ . Let  $x \in X$  and let  $J(x) = \{\alpha \in J : x \in \text{support } \phi_\alpha\}$  be a finite set. Suppose  $\sum_{\alpha \in J(x)} \phi_\alpha(x) = 1$ .  $\{\phi_\alpha\}$  is said to be a partition of unity dominated by  $\mathcal{A}$ .

Claim: let  $X$  be Hausdorff paracompact and  $\mathcal{A}$  be an open covering of  $X$ . There exists a partition of unity  $\{\phi_\alpha\}$  dominated by  $\mathcal{A}$ .

Let  $X$  be Hausdorff paracompact (so  $X$  is normal). Let  $\mathcal{A}$  be an open covering of  $X$ . By regularity (and earlier lemma), pick  $\mathcal{B}$  to be a locally finite open refinement of  $\mathcal{A}$  covering  $X$  such that  $\overline{\mathcal{B}}$  is a refinement of  $\mathcal{A}$ . Pick  $\mathcal{C}$  to be a locally finite closed refinement of  $\mathcal{B}$  covering  $X$ . Index  $\mathcal{C}$  by the set  $J$ .

Let  $\alpha \in J$  and  $C \subset V \subset U$  for  $C \in \mathcal{C}, V \in \mathcal{B}, U \in \mathcal{A}$ . By Urysohn's Lemma, pick  $\phi_\alpha : X \rightarrow I$  such that  $\phi_\alpha(C) = \{1\}$  and  $\phi_\alpha(X - V) = \{0\}$ . Note  $\text{support } \phi_\alpha \subset \overline{V} \subset U$  (we can pick the same  $U$ ). Let  $\mathcal{D} = \{\text{support } \phi_\alpha : \alpha \in J\}$ , so  $\mathcal{D}$  is a locally finite closed refinement of  $\mathcal{A}$ .

Now we may define  $\phi : X \rightarrow \mathbb{R}$  by  $\phi(x) = \sum_{\alpha \in J(x)} \phi_\alpha(x)$ .  $\phi(x) \geq 1$  since  $\mathcal{C}$  covers  $X$ . Let  $y = \phi(x)$ . Since  $\mathcal{D}$  is locally finite, pick  $W(x)$  intersecting finitely many elements of  $\mathcal{D}$ , so over  $W$ ,  $\phi = \phi_1 + \phi_2 + \phi_3 + \cdots + \phi_n$ , which is continuous.

Def: let a function  $f : X \rightarrow Y$  be locally continuous if for all  $x \in X$ , there exists  $U(x)$  such that  $f|U$  is continuous. Claim:  $f$  is continuous iff it is locally continuous. Proof: let  $x \in X$ ,  $y = f(x)$ , and  $V(y)$  be given. Pick  $U(x)$  such that  $f|U$  is continuous. Pick  $W(x) \subset U$  open in  $U$  (and therefore  $X$ ) such that  $f(W) \subset V$ . Then by def,  $f$  is continuous.

$\phi$  is locally continuous, and therefore continuous. Hence,  $\phi'_\alpha = \phi_\alpha / \phi : X \rightarrow I$  is (well-defined), continuous, and  $\phi' = 1$ .

Let  $\{U_\alpha\}_{\alpha \in J}$  be an open cover of  $X$ . Let  $\{V_\beta\}_{\beta \in K}$  be an open refinement of  $\{U_\alpha\}$  covering  $X$ , such that  $\overline{V_\beta} \subset U_\alpha$  for all  $\beta$  and some  $\alpha$ . Pick a function  $f : K \rightarrow J$  such that  $\overline{V_\beta} \subset U_{f(\beta)}$ .

For all  $\alpha \in J$ , define  $W_\alpha = \bigcup_{\beta \in f^{-1}(\alpha)} V_\beta$ , so  $\overline{W_\alpha} \subset U_\alpha$  ( $\{V_\beta\}_{\beta \in f^{-1}(\alpha)}$  is locally finite). I claim  $\{W_\alpha\}_{\alpha \in J}$  is locally finite (this covering is already open and covers  $X$ ).

Let  $x \in X$ . Pick  $N(x)$  intersecting finitely many  $V_\beta$ 's. Suppose  $W_\alpha \cap N \neq \emptyset$ . Then for some  $\beta \in f^{-1}(\alpha)$  we have  $V_\beta \cap N \neq \emptyset$ . If  $W_\alpha \neq W_{\alpha'}$ , then  $V_\beta \neq V_{\beta'}$ , since the fibers are disjoint.

Let  $\{U_\alpha\}$  be an open covering of paracompact Hausdorff  $X$ . Then there exists a partition of unity  $\{\phi_\alpha\}$  dominated by  $\{U_\alpha\}$ .

Pick  $\{V_\alpha\}$  to be a precise locally finite open refinement of  $\{U_\alpha\}$  covering  $X$ , such that  $\overline{V_\alpha} \subset U_\alpha$ . Pick  $\{W_\alpha\}_{\alpha \in J}$  to be a precise locally finite open covering of  $\{V_\alpha\}$  having the same property. Hence, we have constructed  $\mathcal{C} < \mathcal{B} < \mathcal{A}$ , where  $<$  denotes refinement, each covering is of  $X$ , and all collections are indexed by  $J$ . Then proceed as before.

Let  $X$  be paracompact Hausdorff. Let  $\mathcal{C}$  be a locally finite collection of subsets of  $X$ . For each  $C \in \mathcal{C}$ , assign  $\epsilon_C > 0$ . There exists a function  $f : X \rightarrow \mathbb{R}^+$  such that if  $x \in C$ , then  $f(x) \leq \epsilon_C$ .

Let  $X$  be indexed by  $J$ . Let  $x \in X$  and  $U_\alpha(x)$  intersect finitely many ele-

ments of  $\mathcal{C}$ , which form a subcollection  $\mathcal{D}$ . If  $\mathcal{D} \neq \emptyset$ , let  $\epsilon_\alpha = \min_{D \in \mathcal{D}} \epsilon_D > 0$ . Else,  $\epsilon_\alpha = 1$ . Let  $\{\phi_\alpha\}$  be a partition of unity dominated by  $\{U_\alpha\}$ . Define  $\phi'_\alpha = \epsilon_\alpha \phi_\alpha$ . I claim  $\phi' : X \rightarrow \mathbb{R}^+$  given by  $\phi'(x) = \min_{x \in \phi_\alpha^{-1}(\mathbb{R}^+)} \phi'_\alpha(x)$  is our desired function.

Since  $\{\phi_\alpha^{-1}(\mathbb{R}^+)\}$  is locally finite, it is also point-wise finite, so  $\phi'(x) > 0$ . If  $x \in X$ , pick  $V(x)$  intersecting finitely many elements of  $\{\phi_\alpha^{-1}(\mathbb{R}^+)\}$ , say which are indexed by  $K \neq \emptyset$ . Then  $\phi'|_V = \min_{\alpha \in K} \phi'_\alpha$  is continuous, so  $\phi'$  is continuous. If  $x \in C$ , then  $x \in \phi_\alpha^{-1}(\mathbb{R}^+) \subset U_\alpha$  (for some  $\alpha$  not necessarily associated with  $x$ ), so  $U_\alpha \cap C \neq \emptyset$ , so  $\phi'(x) \leq \phi'_\alpha(x) \leq \epsilon_\alpha \leq \epsilon_C$ .

Show that the product of a paracompact space  $X$  with a compact space  $Y$  is paracompact.

Let  $\mathcal{A}$  be a basic covering of  $X \times Y$ . Since  $Y$  is compact, let  $x \in X$  and cover  $\{x\} \times Y$  with a finite subcollection  $\mathcal{A}_x$ . (Essentially by Tube lemma)  $U(x) = \bigcap \pi_1(\mathcal{A}_x)$  is open and  $\mathcal{B}_x = \pi_2(\mathcal{A}_x)$  covers  $Y$ .

The set  $\{U\}$  covers  $X$ . Since  $X$  is paracompact, pick locally finite open refinement  $\mathcal{V} = \{V\}$  covering  $X$ . Let  $f : \mathcal{V} \rightarrow X$  be a function such that  $V \subset U(f(V))$ . Let  $\mathcal{A}' = \bigcup_{V \in \mathcal{V}} \{V \times W : W \in \mathcal{B}_{f(V)}\}$ . I claim  $\mathcal{A}'$  is a locally finite open refinement of  $\mathcal{A}$  covering  $X$ .

Sps  $(x, y) \in X \times Y$ . Pick  $V \in \mathcal{V}$  st  $x \in V$ . Pick  $W \in \mathcal{B}_{f(V)}$  st  $y \in W$ . Then  $(x, y) \in V \times W \in \mathcal{A}'$ . Sps  $(x, y) \in X \times Y$ . Pick  $N(x)$  intersecting finitely many elements of  $\mathcal{V}$ . Then  $N \times Y$  intersects finitely many elements of  $\mathcal{A}'$  (each  $\mathcal{B}_{f(V)}$  is finite). Lastly,  $\mathcal{A}'$  is a refinement of  $\mathcal{A}$ .

$\omega_1$  is not paracompact. If otherwise,  $\omega_1 \times (\omega_1 + 1)$  is paracompact ( $\omega_1 + 1$  is compact) Hausdorff, cont as  $\omega_1 \times (\omega_1 + 1)$  is not normal.  $\omega_1$  is locally compact, but not paracompact.

Let  $X$  be paracompact. It does not follow that every open covering  $\mathcal{A}$  of  $X$  has a locally finite subcollection  $\mathcal{A}'$  covering  $X$ . Let  $X = \mathbb{R}$ , paracompact since metrizable. Let  $\mathcal{A} = \{(-n, n) : n \in \mathbb{N}\}$ . Sps FTSOC there exists  $\mathcal{A}' \subset \mathcal{A}$  locally finite covering  $X$ . In particular,  $0 \in \bigcup \mathcal{A}'$ , so finitely many elements of  $\mathcal{A}'$  contain 0. Hence,  $\mathcal{A}'$  is finite, so  $\bigcup \mathcal{A}' \neq X$ , cont.

Let  $X = \omega_1$  have the discrete topology (so  $X$  is metrizable and thus paracompact). Then  $i : X \rightarrow \omega_1$  is cont, but  $f(X) = \omega_1$  is not paracompact.

Let  $X = A \cup B$  be compact Hausdorff,  $A, B$  closed metrizable. Then  $X$  is metrizable.

$X$  is regular (in fact normal), so ISTS  $X$  is second-countable.  $A, B$  are both compact metrizable and therefore second-countable. Let  $\mathcal{B}_A, \mathcal{B}_B$  be count-



able bases for  $A, B$ , resp. Let  $x \in A \cap B$ . Let  $B_A(x, 1/n) = U_A \cap A$  and  $B_B(x, 1/n) = U_B \cap B$ , so  $U = U_A \cap U_B$  is an nhod of  $x$ . Since  $\{U\}$  covers  $A \cap B$  and  $A \cap B$  is compact, let  $\mathcal{B}_n = \{U'\}$  be a finite subcovering. Let  $\mathcal{B}_M = \bigcup_{n < \omega} \mathcal{B}_n$  be a countable covering of  $A \cap B$ .

I claim  $\mathcal{B} = (\mathcal{B}_A \cap (X - B)) \cup \mathcal{B}_M \cup (\mathcal{B}_B \cap (X - A))$  is a basis for  $X$  (it is countable). If  $x \notin B$  and  $U(x)$  is open, pick  $N \in \mathcal{B}_A$  such that  $x \in N \subset U \cap A \subset U$ .  $N = V \cap A$  for some  $V$  open in  $X$ , so  $N \cap (X - B) = V \cap (X - B)$  is open.

If  $x \in A \cap B$ , and  $U(x)$  is open, pick  $n$  st  $B_A(x, 1/n) \subset U \cap A$  and  $B_B(x, 1/n) \subset U \cap B$ . Pick  $m > 2n$  and  $W = U_A \cap U_B \in \mathcal{B}_m$  containing  $x$ . ISTS  $W \cap A \subset U \cap A$  and  $W \cap B \subset U \cap B$ .  $W \cap A = U_A \cap U_B \cap A = B_A(y, 1/(2m)) \cap U_B \subset B_A(x, 1/n) \subset U \cap A$ .

Let  $\mathcal{A} = \bigcup_{n < \omega} \mathcal{A}_n$  a countably locally finite open covering of  $X$ . There exists a locally finite refinement (not necessarily open nor closed)  $\mathcal{A}'$  of  $\mathcal{A}$  covering  $X$ .

Let  $X_n = \bigcup \mathcal{A}_n$  and  $S_n(A) = A - \bigcup_{k < n} X_k$ . Let  $\mathcal{A}'_n = S_n(\mathcal{A}_n)$ . I claim  $\mathcal{A}' = \bigcup_{n < \omega} \mathcal{A}'_n$  is our desired covering. First,  $S_n(A) \subset A$ , so  $\mathcal{A}'$  is a refinement. Suppose  $x \in X$ . Take  $x \in X_n$  with  $n$  minimal. Then  $x \in X_n - \bigcup_{k < n} X_k$ . Pick  $x \in A \in \mathcal{A}_n$ , so  $x \in A - \bigcup_{k < n} X_k = S_n(A)$ .

Let  $X'_n = \bigcup \mathcal{A}'_n$ , so  $X = \bigcup_{n < \omega} X'_n$ . Suppose  $x \in X'_n$ . Since each  $\mathcal{A}'_k$  is locally finite, pick for all  $k \leq n$  pick  $N_k(x)$  intersecting finitely many elements of  $\mathcal{A}'_k$ . Note  $x \in A \in \mathcal{A}_n$ . Let  $N = A \cap \left( \bigcap_{k \leq n} N_k \right)$  be an nhod of  $x$ , which intersects finitely many elements of  $\bigcup_{k \leq n} \mathcal{A}'_k$ . For  $m > n$ ,  $S_m(B) \cap N = \emptyset$  since  $N \subset A \subset X_n$ .

Let  $X$  be paracompact. Let  $\mathcal{B} = \{B_\alpha\}_{\alpha \in J}$  be a locally finite collection of subsets of  $X$ . There exists a locally finite open collection  $\mathcal{A} = \{A_\alpha\}_{\alpha \in J}$  such that  $B_\alpha \subset A_\alpha$  for all  $\alpha \in J$ .

For all  $x \in X$ , pick  $N(x)$  intersecting finitely many elements of  $\mathcal{B}$ . By paracompactness, let  $\mathcal{A}'$  be a locally finite open refinement of  $\{N\}$  covering  $X$ . For all  $B \in \mathcal{B}$ , define  $U(B) = \bigcup \{A' \in \mathcal{A}' : A' \cap B \neq \emptyset\}$ . I claim the set of  $U(B)$ 's is our desired collection  $\mathcal{A}$ .

First, for any  $x \in B$ , we may pick  $x \in A'$  since  $\mathcal{A}'$  covers  $X$ . Then  $B \cap A' \neq \emptyset$ , so  $x \in A' \subset U(B)$ . That is,  $B \subset U(B)$ . Note  $U(B)$  is open.

Since  $\mathcal{A}'$  is locally finite, for  $x \in X$  pick  $M(x)$  intersecting finitely many elements of  $\mathcal{A}'$ . Since  $\mathcal{A}'$  covers  $X$ ,  $M$  is covered by finitely many elements of  $\mathcal{A}'$ , say  $\mathcal{D}$ . Each  $A' \in \mathcal{A}'$  intersects finitely many  $B$ 's, being a refinement of  $\{N\}$ .

Suppose  $U(B) \cap M \neq \emptyset$ . Then  $A' \cap M \neq \emptyset$  and  $A' \cap B \neq \emptyset$  (by construction of  $U(B)$ ).  $A' \in \mathcal{D}$ . Hence, there are only finitely many  $B$ 's for which  $U(B) \cap M \neq \emptyset$ .

Let  $X = A \cup B$  be regular with  $A, B$  closed paracompact. Then  $X$  is paracompact.

(a). Let  $\mathcal{A}$  be an open cover of  $X$ . Then  $\mathcal{A} \cap A$  is an open cover of  $A$ . Since  $A$  is regular paracompact,  $\mathcal{A} \cap A$  has a locally finite closed refinement  $\mathcal{B}_A$  covering  $A$ . Then  $\mathcal{B}_A$  is locally finite closed wrt  $X$ . Similarly, there exists locally finite closed wrt  $X$   $\mathcal{B}_B$  covering  $B$ , so  $\mathcal{B}_A \cup \mathcal{B}_B$  is a locally finite closed covering of  $X$  refining  $\mathcal{A}$ . Since  $X$  is regular and every open covering  $\mathcal{A}$  has a locally finite closed refinement covering  $X$ ,  $X$  is paracompact.

(b). Suppose  $X = \bigcup_{n < \omega} X_n$ . Let  $\mathcal{A}$  be an open covering of  $X$ . For each  $n \in N$ , since  $X_n$  is paracompact, pick a locally finite open (wrt  $X_n$ ) refinement  $\mathcal{A}_n$  of  $\mathcal{A} \cap X_n$  covering  $X_n$ . Let  $\mathcal{A}'_n = \mathcal{A}_n \cap X_n^\circ$  be a locally finite open (wrt  $X$ ) refinement covering  $X_n^\circ$ . Let  $\mathcal{A}' = \bigcup_{n < \omega} \mathcal{A}'_n$ . If  $x \in X$ , then  $x \in X_n^\circ$  for some  $n$ , so  $x \in \bigcup \mathcal{A}'_n$ , i.e.,  $\mathcal{A}'$  is an open covering of  $X$ .

Since  $X$  is regular and any open covering  $\mathcal{A}$  has a countably locally finite open refinement  $\mathcal{A}'$ ,  $X$  is paracompact.

Let  $X$  be locally compact Hausdorff and  $A \subset X$  be closed. Let  $x \in A$  such that  $x \in U \subset C$ . Then  $x \in U \cap A \subset C \cap A$ .  $C \cap A$  is a closed subset of the compact space  $C$ , so  $C \cap A$  is compact. Hence,  $A$  is locally compact.

If  $C \subset X$  is a connected component, so is  $\overline{C}$ , so  $C = \overline{C}$  is closed.

Let  $X$  be locally compact, paracompact, connected, Hausdorff. Then  $X$  is second-countable?

$X$  is normal, since it is paracompact Hausdorff. Let  $\{U\}$  be a covering of  $X$  with open metrizable subspaces. By paracompactness, pick locally finite open refinement  $\mathcal{V} = \{V\}$  covering  $X$ . Since each  $V$  is metrizable, pick countably locally finite basis  $\mathcal{B}_V = \bigcup_{n < \omega} \mathcal{B}_{Vn}$ . Let  $\mathcal{B} = \bigcup_{V \in \mathcal{V}} \mathcal{B}_V$  (consisting of open sets in  $X$ ).

I claim  $\mathcal{B}$  is a countably locally finite basis for  $X$ . Let  $\mathcal{B}_n = \bigcup_{V \in \mathcal{V}} \mathcal{B}_{Vn}$ . Let  $x \in X$ . By local finiteness of  $\mathcal{V}$ , pick  $N(x)$  intersecting finitely many elements of  $\mathcal{V}$ , say  $V_1, V_2, \dots, V_n$ . By local finiteness of  $\mathcal{B}_{V_i n}$ , pick  $N_i(x)$  intersecting finitely many elements of  $\mathcal{B}_{V_i n}$ . Let  $W(x) = N \cap \left( \bigcap_{i \in [n]} N_i \right)$ , so  $W$  intersects finitely many elements of  $\mathcal{B}_n$ .

Further, if  $x \in X$ , pick  $x \in V$ . For  $x \in M$  open,  $x \in V \cap M$ . Pick  $B_{Vn} \in \mathcal{B}$  st

$x \in B_{V_n} \subset V \cap M$ , so  $\mathcal{B}$  is indeed a basis.

Let  $p : X \rightarrow Y$  be a perfect map,  $Y$  paracompact. Then  $X$  is paracompact?